SKILL BUILDERS · UNIT 6 · SCORE 3

Functions: Practice Set

NC.8.F.1–5 · NC.8.EE.8

Name _____ Date _____ Period _____

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.F.1

$f(x) = 4x + 1$. Find $f(2)$.

Answer: _____

2 Score 3 NC.8.F.1

Which table shows a function?

(A)

x	1	1	2	3
y	4	5	6	7

(B)

x	2	3	2	4
y	1	2	3	4

(C)

x	1	2	3	4
y	5	5	6	6

3 Score 3 NC.8.F.4

Find the slope from **(1, 3)** to **(4, 12)**.

Answer: _____

4 Score 3 NC.8.F.4

Find the slope from **(0, 6)** to **(3, 0)**.

Answer: _____

5 Score 3 NC.8.F.4

Which equation matches the table?

x	0	1	2
y	5	8	11

(A) y = 3x

(B) y = 3x + 5

(C) y = 5x + 3

(D) y = x + 5

6 Score 3 NC.8.F.2

A savings account has **y = 5x + 20** dollars after x weeks. What is the **start value**?

Answer: _____ dollars

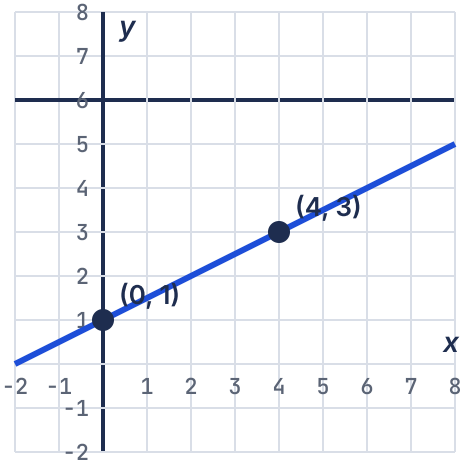
7 Score 3 NC.8.F.2

In **y = 5x + 20**, what is the **rate** (amount added each week)?

Answer: _____ dollars

8 Score 3 NC.8.F.4

What is the slope of this line?



Answer: _____

9 Score 3 NC.8.EE.8

Solve the system: **y = x + 2** and **y = 3x**.

(A) (1, 3)

(B) (2, 4)

(C) (3, 1)

(D) (0, 2)

10 Score 3 NC.8.F.5

A runner's distance-from-home graph is **flat** for a while. What is the runner doing?

(A) Stopped: the distance is not changing

(B) Running faster

(C) Running home

11 Score 3 NC.8.F.4

Find the mistake. For (2, 1) and (4, 7), Kai wrote: `slope = (4 - 2) ÷ (7 - 1) = 2 ÷ 6 = 1/3`. What went wrong?

(A) Kai put the run on top. Slope = rise ÷ run = 6 ÷ 2 = 3.

(B) Kai is right.

(C) The slope should be negative.

(D) Kai should add: (4 + 2) ÷ (7 + 1).